

A220 – The Tolerance Standard: Four Precision Classes

Johann Pascher

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.1 Three Numbers, Not One

[STIPULATION] For every comparison number the series gives three quantities, never one: the value, the measurement precision of the comparison quantity, and the determination tolerance of the theory quantity. A percentage deviation alone says nothing.

The reason is simple and used throughout: a one per cent deviation is meaningless when the theory quantity is only determined to two significant figures – and damning if both sides are fixed to eight figures. Without both tolerances the percentage is unreadable.

.2 We Compute Nothing

[STIPULATION] The most important statement about all numbers in this series: the theory *computes nothing*. It is not a calculation engine producing predictions to many decimal places. It orders geometric structures and shows that they fall in the right range.

From this follows how the verification scripts are to be read. When a script reports agreement “to 10^{-15} ”, that is *not* a result. It is either an **identity** – a quantity equal by definition or construction ($\alpha = 1$ after charge normalisation, $L_0/\ell_P = \xi$ by definition) – or it is a **range check**: the test of whether we are in the right order of magnitude. The many decimal places are floating-point arithmetic, not the precision of a statement.

[B] Three types of test, strictly to be distinguished:

An *identity* verifies a definition. It cannot fail as long as the algebra is correct. Passing it is no achievement; failing it would be a programming error. Examples: $\alpha = 4\pi/4\pi = 1$ (A130), $L_0 = \xi \ell_P$ (A210), the reverse direction $\xi = 2\sqrt{Gm}$ (A145).

A *range check* compares a structure with measurement and asks only: correct order of magnitude, correct sign, correct scaling? Whether E_{char} is near 28 rather than 3 or 300, whether κ is near 4 rather than 9, whether the CHSH residual is at 10^{-4} rather than 10^{-1} .

A *prediction* is the case where the theory *genuinely* predicts something measurable: the **lepton masses**. The structure $(\sqrt{2}, 2/9)$ fixes m_τ from m_e and m_μ – a sharp, falsifiable prediction, $m_\tau = 1776.968$ MeV against PDG 1776.86 ± 0.12 , i.e. $+0.90\sigma$. This is not a range check: here a numerical value stands *before* the measurement, and the measurement decides it.

[STIPULATION] “We compute nothing” therefore applies to the *bookkeeping values* – $\alpha = 1/137$, the SI value of G , the anchor choice L_0 – but not to the lepton masses. For the bookkeeping values the question “is the result exactly correct?” is wrongly posed; for m_τ it is exactly right. The difference: the lepton masses are bound so tightly by a *ratio* structure that the third mass follows from two – that is a genuine prediction, not an absolute value and not a calibration.

.3 The Four Precision Classes of the Series

[B] Across the whole series four classes appear. Distinguishing them is the actual work of the tolerance standard.

Class 1 – five figures and more. The circulant (A110): $Q = 2/3$ to six figures, $\theta(\mu/e)$ to seven. Here a deviation of 10^{-6} is a serious signal, because both sides are fixed that precisely.

Class 2 – per cent level. The ladder (A100): μ/e to 0.52 %, τ/μ to 1.04 %. Here one per cent is *not* a signal against the structure, because the structure claims nothing more precise at this level – and no success that one could report to five figures.

Class 3 – bookkeeping precision. The SI value of α (A130, +0.0005 % with $E_0 = 7.398$ MeV; Route 2: $\sqrt{m_e m_\mu / K_{\text{frak}}} = 7.3974$ MeV, –0.017 %) and of G (A145; E_{char} cancels in G , the SI value hits +0.003 %). On the precision of G : C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383, [K]); with the anchor m_e , G lies at –2.6 %, with ν at –0.46 % (Doc. 383). The +0.003 % arises only with this residual and does not count as an independent precision result (Doc. 190, R141). These deviations concern the scale conversion, not the structure. They are not a test (A210); the percentage measures the bridge, not the theory.

Class 4 – undetermined. The neutrino sector (A150), the quark masses (A150, scheme-dependent). Here the theory quantity is so little fixed that any agreement is inconsequential. What cannot be refuted confirms nothing.

.4 The Most Common Errors

[B] The tolerance standard exists to avoid five errors that all occurred at least once in the corpus:

Confusing classes. Taking a class-2 deviation (ladder, per cent) against a class-1 precision (circulant, five figures) – that is the level error from A080/A110. Both describe the same masses, but at different levels.

Reading bookkeeping as a test. Counting a class-3 deviation as a confirmation or refutation. The error A130 names in the α presentation.

Counting undetermined as confirmed. Booking a class-4 quantity as a success because the theory “lies in the range” – even though the range is so wide that anything fits.

Inventing figures. Giving a class-2 quantity five significant figures because the calculation provides them. The floating-point precision of a calculation is not the determination tolerance of the quantity.

Counting dependent as independent. Presenting a quantity as independent evidence or as a second figure, even though it *follows arithmetically* from one already counted. Example from the ladder (A100): in the ratio $N_1:N_2:N_3 \approx 1:1.981:2.971$, only $N_2 = 1.981$ is measured – the third number is forced as $1 + 1.981$ and carries no independent evidence.

The standard requires booking such forced figures for what they are – consequence, not finding.

.5 The Standard in One Sentence

[STIPULATION] An agreement counts only as far as the *worse* determined of the two sides reaches. A deviation weighs only as heavily as the *better* determined of the two sides supports it.

That is the whole standard. Everything else is its application.

.6 Verification Script

- Classification of the comparison numbers of the series into the four classes, with presentation of measurement precision and determination tolerance per case; and the countercheck that a class-2 number reported to five figures generates spurious precision: `python/A_Serie_Skripte/a220_toleranzmassstab.py`

.7 Open Questions

First, the determination tolerance of some theory quantities is itself not sharp. For the ladder exponents (A100) it is unclear to how many figures they are determined at all. Where that is the case, the standard is to be applied conservatively.

Second, the standard does not replace error propagation. It orders orders of magnitude; it does not propagate uncertainties through formulas. A full error propagation the series does not provide.

.8 Supersedes

This document makes the tolerance rule from A010 and A080 into its own principle and applies it to all sectors. It subsumes the corresponding parts of Doc. 122 (ratio and absolute) and Doc. 292 (tolerance analysis leptons). Incorporated: R55 (evidential breadth of the ladder coupling: one observation instead of three; "factor 300" not as a figure).

.9 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.